

Radioactive Decay and Half-Life

Nuclear equations, radiation and half-life

Name _____ Class _____ Date _____

RULES AND DATA

Alpha ${}^4_2\text{He}$: $A - 4$, $Z - 2$ · Beta ${}^0_{-1}\text{e}$: $Z + 1$ · Positron ${}^0_{+1}\text{e}$: $Z - 1$ · Gamma: no change

$$N = N_0 \times \left(\frac{1}{2}\right)^n, n = t \div t_{1/2} \cdot t = t_{1/2} \times \ln(N_0/N) \div \ln 2$$

Z: N 7 · O 8 · Ca 20 · Ni 28 · Xe 54 · Rn 86 · Th 90 · Pa 91 · U 92

Use the same time unit for t and $t_{1/2}$. Show units on every quantity.

A. Understanding

1. Define (a) radioactivity (b) half-life. (c) Why can we predict the half-life of a large sample but not the moment when one particular nucleus will decay?

B. Nuclei and nuclear equations

2. Give the numbers of protons and neutrons in: (a) ${}^{235}_{92}\text{U}$ (b) ${}^{60}_{27}\text{Co}$ (c) ${}^{131}_{53}\text{I}$ (d) ${}^{222}_{86}\text{Rn}$

3. Complete each equation: (a) ${}^{226}_{88}\text{Ra} \longrightarrow ? + {}^4_2\text{He}$ (b) ${}^{131}_{53}\text{I} \longrightarrow ? + {}^0_{-1}\text{e}$ (c) ${}^{18}_9\text{F} \longrightarrow ? + {}^0_{+1}\text{e}$ (d) ${}^{60}_{27}\text{Co} \longrightarrow ? + {}^0_{-1}\text{e}$ (e) ${}^{239}_{94}\text{Pu} \longrightarrow ? + {}^4_2\text{He}$ (f) ${}^{40}_{19}\text{K} \longrightarrow ? + {}^0_{-1}\text{e}$

4. Identify X: (a) ${}^{234}_{90}\text{Th} \longrightarrow {}^{234}_{91}\text{Pa} + \text{X}$ (b) ${}^{210}_{84}\text{Po} \longrightarrow {}^{206}_{82}\text{Pb} + \text{X}$ (c) ${}^{11}_6\text{C} \longrightarrow {}^{11}_5\text{B} + \text{X}$

5. Uranium-238 (${}^{238}_{92}\text{U}$) decays by alpha emission, then beta, then beta again. Write the three equations and identify the final nucleus.

C. Types of radiation

6. Compare alpha, beta and gamma radiation: what each one is, its charge, and what stops it. Explain why an alpha emitter is most dangerous inside the body.

D. Half-life

7. A patient is given technetium-99m with an activity of 800 MBq ($t_{1/2} = 6.01 \text{ h}$). What is its activity after 24.04 h?

8. (a) The activity of a sample falls from 1600 Bq to 400 Bq in 30.0 min. Find its half-life. (b) Radon-222 has $t_{1/2} = 3.82$ d. What percentage of a sample remains after 10.0 d?

E. Challenge

9. A sample of carbon from an ancient wooden bowl has a carbon-14 activity of 0.0850 Bq per gram of carbon. Carbon from fresh wood measured in the same laboratory gives 0.230 Bq per gram. Estimate the age of the bowl ($t_{1/2}$ of carbon-14 = 5730 yr).